Drug Design in the era of Machine Learning and Computational Intelligence

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The Drug Development Process

But, what’s a drug?

A chemical or biological substance that, when administered to a living organism, interacts with one or more biological targets, producing a therapeutic, preventive, or diagnostic effect by modifying physiological or pathological functions.

Biological targets:

- Proteins (>90% — e.g., enzymes involved in cellular signaling)
- Nucleic acids (e.g., regulation of gene expression, antivirals)
- Non-conventional targets (e.g., macromolecular complexes)

Drug Design

Identification of molecules with biological activity

Optimization of affinity and selectivity to a target

Optimization of ADMET properties

Integration of chemical, biological and pharmacological requirements

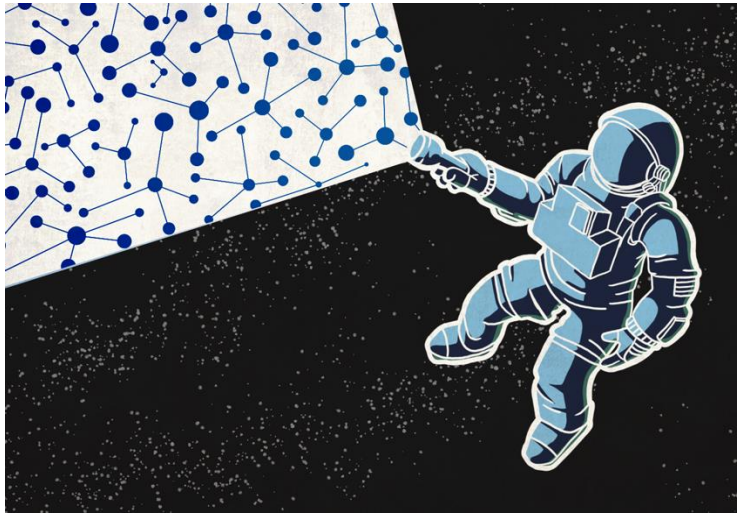
Structure-Based Drug Design (SBDD)
Based on target 3D structure (X-ray, NMR, cryo-EM, AlphaFold)

Ligand-Based Drug Design (LBDD)
Based on known ligands and structure-activity relations (QSAR, pharmacophores)

***De novo* Drug Design**
De novo generation of molecules compatible with the active site

Fragment-Based Drug Design (FBDD)
Assembly/optimization of low-affinity small fragments

Chemical space



Set of all possible chemical compounds → the more it is explored:

- > probability of identifying biologically active molecules
- > chances of finding candidates meeting ket paramenters for clinical advancement

BUT ~10^{60} molecules vs ~10^{24} stars in the observable universe. Impossible to fully explore or screen!

We can accelerate and optimize the exploration with computational techniques.

Computer-Aided Drug Design

Predicting Metabolic Reactions with a Molecular Transformer

Optimization of Cyclic Peptide Binders with Structure-Guided Design Approach

Evolution of High-Affinity and Synthesizable Molecules with a Genetic Algorithm

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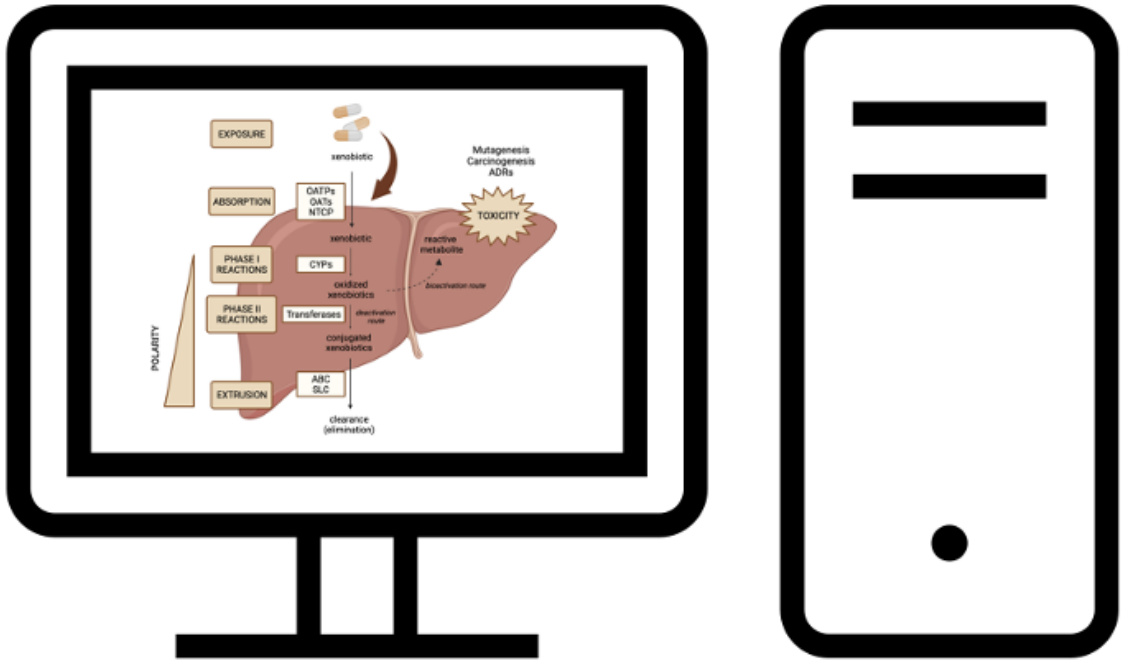


Metabolism is mainly mediated by the liver.

It makes the drug more hydrophilic to be excreted.

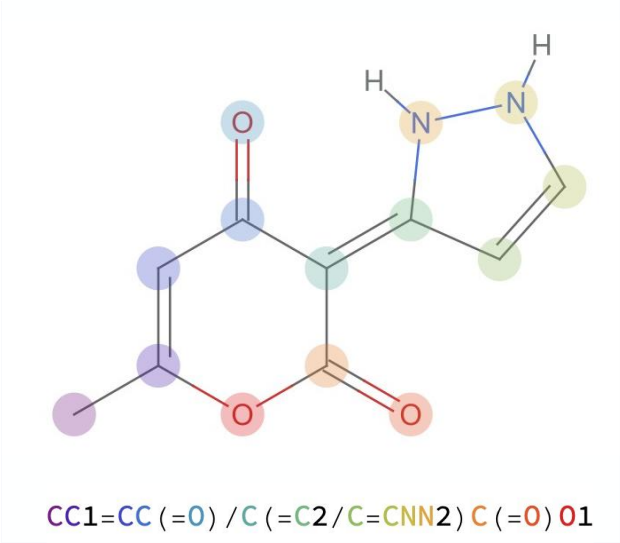
Influences drugs therapeutic effect (and their toxicity).

It is a key topic in medicinal chemistry.

BUT poorly investigated.

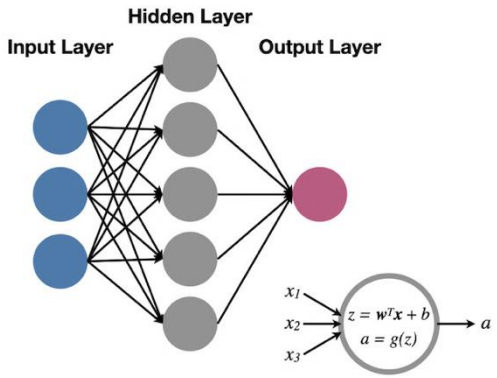
AIM: leveraging the ability of deep learning to automatically learn directly from the data.

Predicting metabolic reactions with a Molecular Transformer

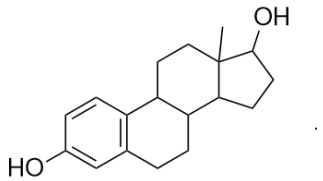


Simplified Molecular-Input Line-Entry System (SMILES)

CC12CCC3C(CCc4cc(O)ccc43)C1CCC2O

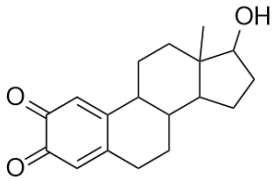


CC12CCC3C(CCC4=CC(=O)C(=O)C=C43)C1CCC2O



ESTROGEN

TRANSFORMER

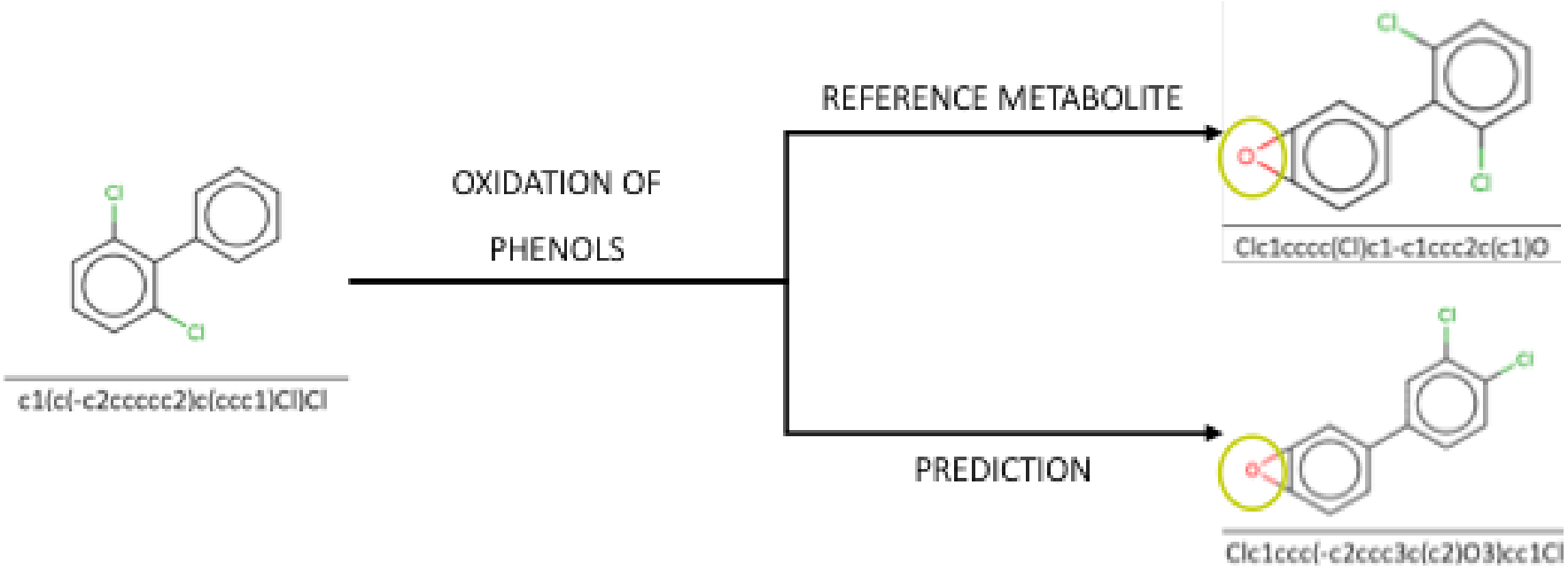


ALGESTONE

Metabolism prediction is casted as a translation problem: substrates are directly converted into the metabolites.

Predicting metabolic reactions with a Molecular Transformer

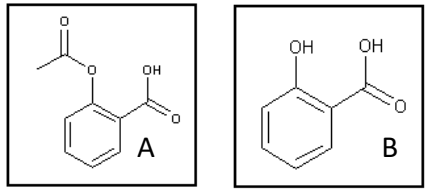
The model is aware of the reaction ID, and is capable of predicting both the site of metabolism and what the reaction is.

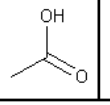
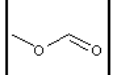


Predicting metabolic reactions with a Molecular Transformer

Correctness of the predictions assessed using fingerprint similarity based on the **Tanimoto coefficient** (S) computed on **extended connectivity fingerprints** (ECFPs).

$$S = \frac{c}{a + b - c}$$

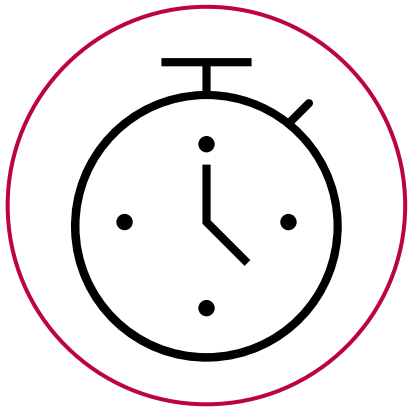


A	1	1	0	1	1	0	1	0
B	1	1	0	1	0	0	0	0
								

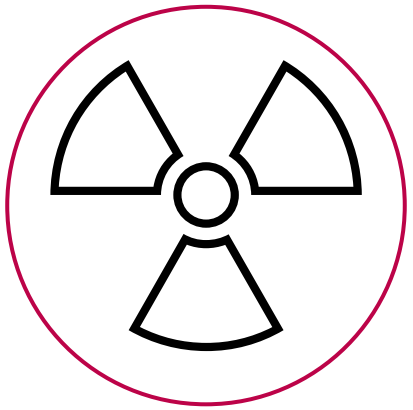
Metric	Model 1	**Model 2**	MetaTrans
Recall	**99.6**	**99.6**	57.9
Precision	54.5	**62.4**	53.4
F1 score	70.5	**76.8**	55.5
Accuracy	54.1	**62.3**	38.8

Predicting metabolic reactions with a Molecular Transformer

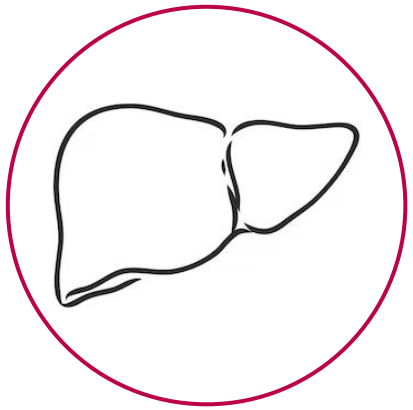
CONCLUSIONS



Prioritizing most promising candidates.



Identification of potential toxicities and drug-drug interactions.



Understanding of the underlying biological mechanisms.

FUTURE PLAN: Automatization of the analysis of the predictions

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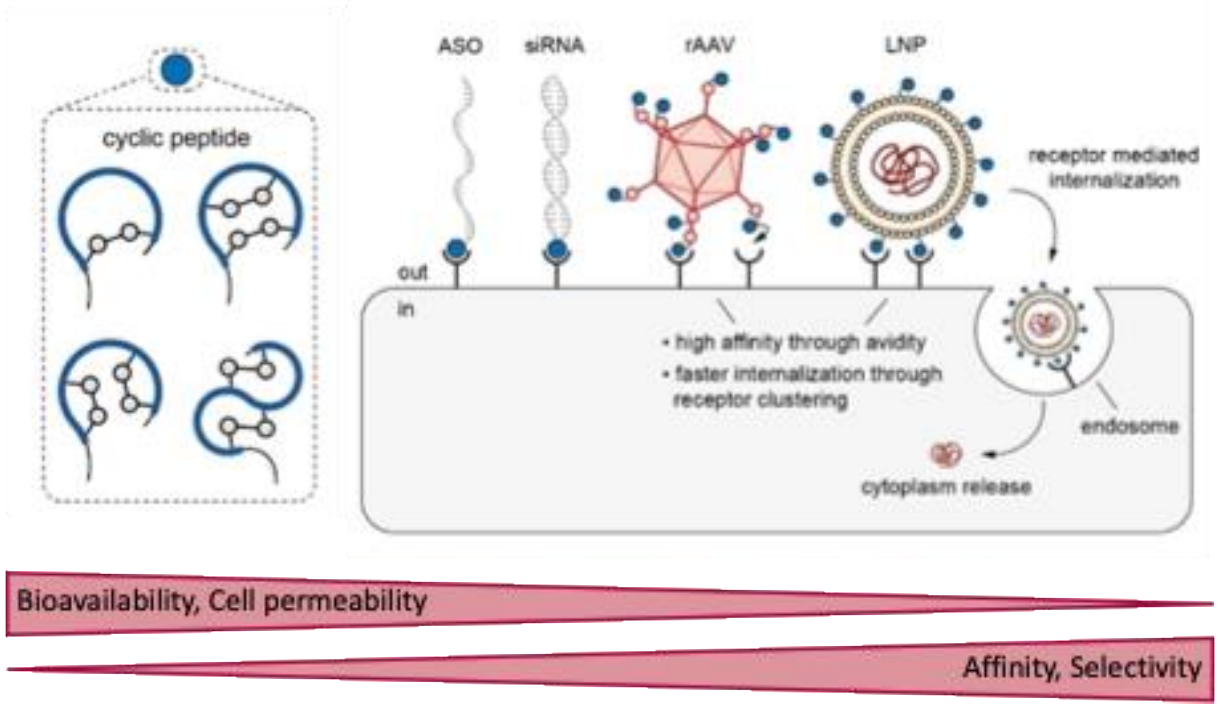


Nucleic Acids-based Terapeuthics:

- Have the potential to treat genetic diseases.
- Safe, targeted, and efficient delivery requires nanocarriers (vs degradation, cell barriers, immunogenicity).

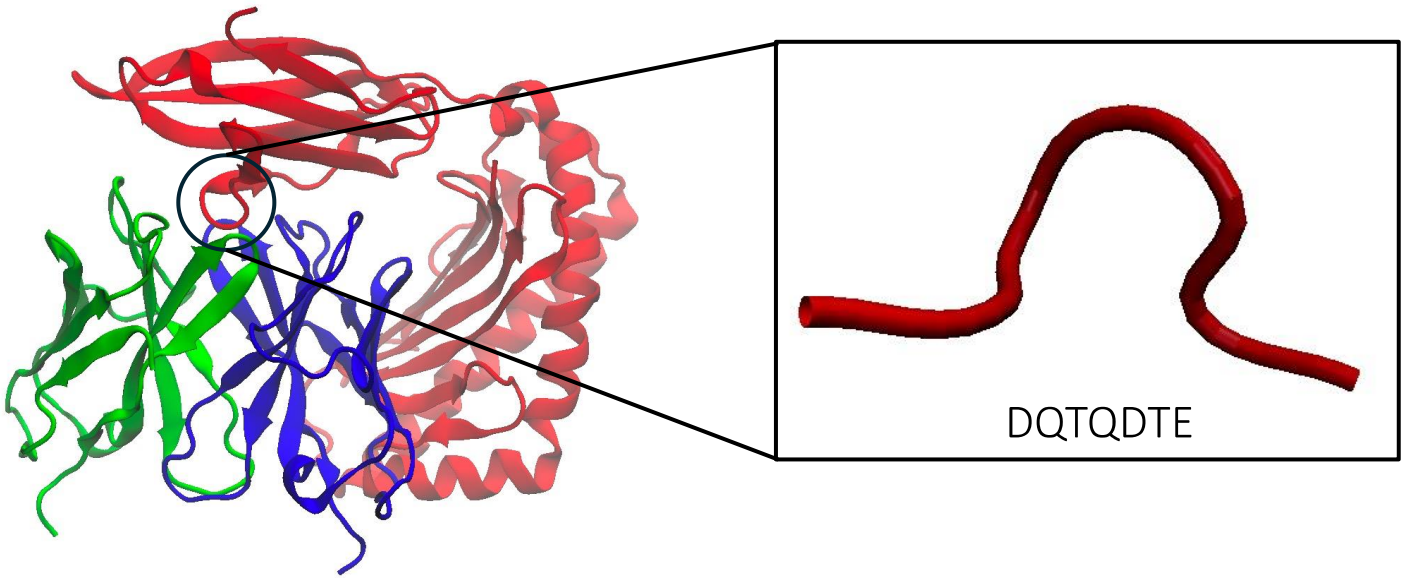
Cyclic peptides: emerging as ideal nanocarriers.

Goal: computational design of cyclic peptides targeting tissue-specific receptors.

Optimization of existing peptide binders

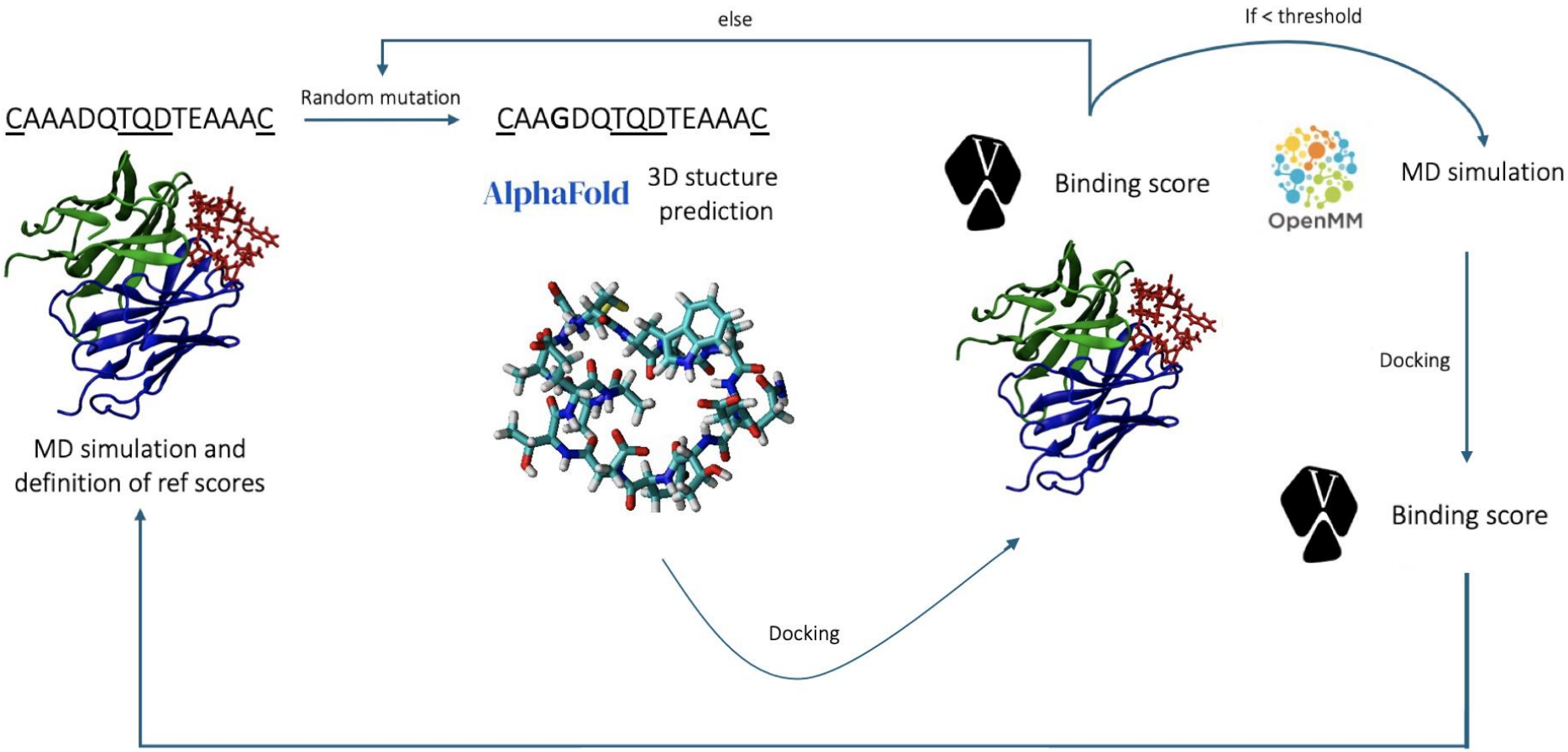
MOLECULAR MODELING

TARGET	KNOWN MOTIF (and derivation)	PDB ID
CD8	DQTQDTE (α3 domain of a human Class I MHC molecule)	1AKJ



Optimization of existing peptide binders

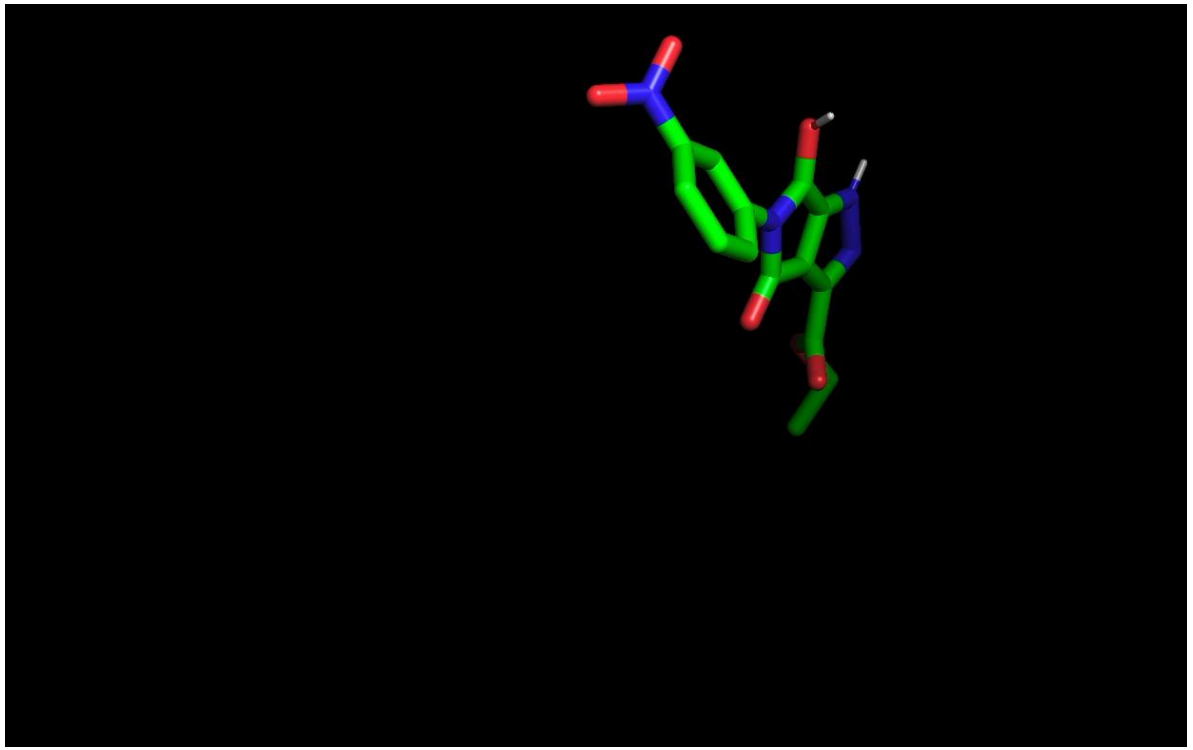
MONTE CARLO ALOGRITHM



For each score: new_score – old_score
If delta_score < 0: Mutation accepted
Else: Metropolis rule to accept/reject

Optimization of existing peptide binders

MOLECULAR DOCKING



Explores possible orientations and conformations in the binding site, generated based on geometric and chemical complementarity.

The interaction is evaluated using scoring functions that approximate binding affinity.

It identifies plausible binding modes, but NOT full system flexibility or time evolution.

Role in the workflow: rapid generation and ranking of peptide–target binding hypotheses.

Optimization of existing peptide binders

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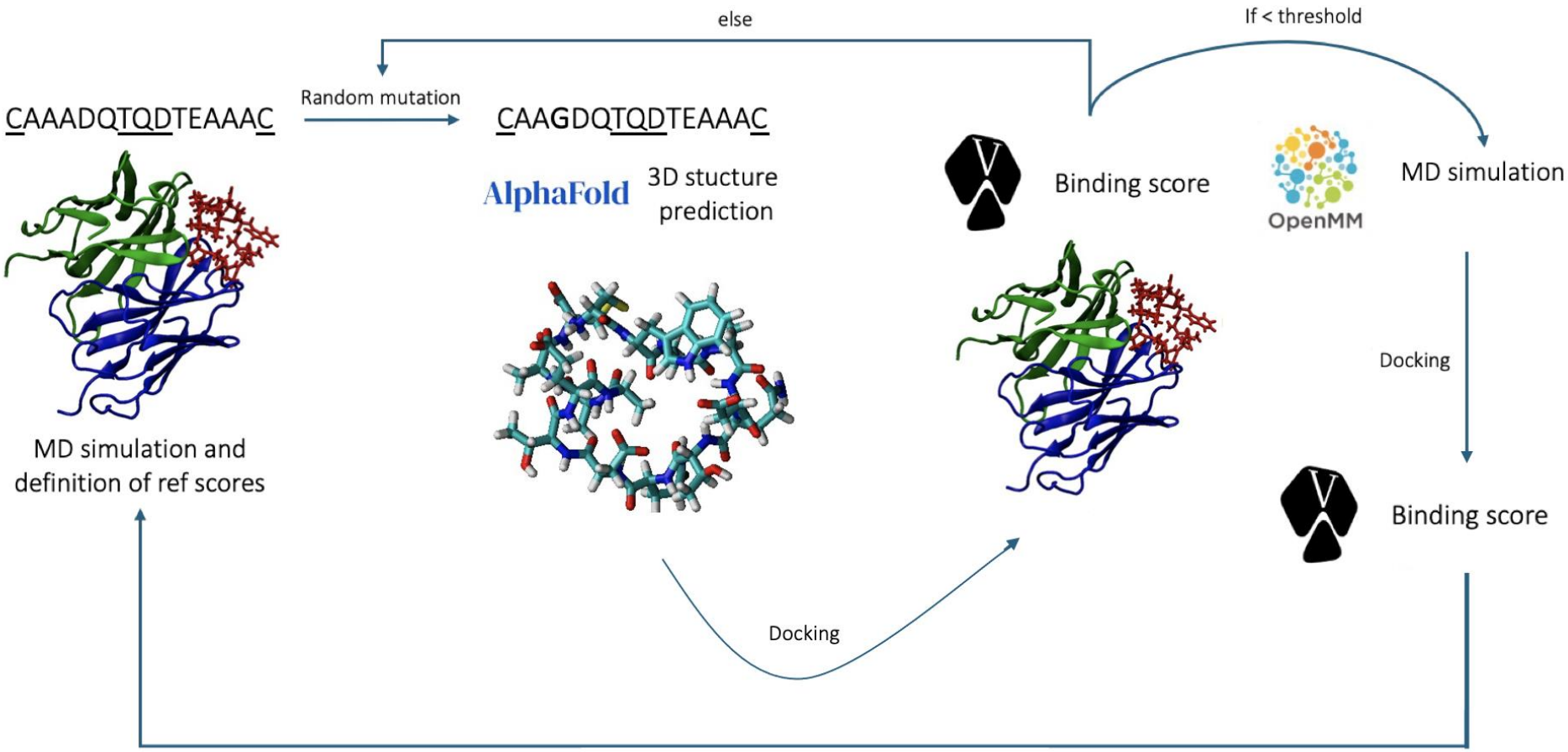

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Optimization of existing peptide binders

MOLECULAR DYNAMICS

The system is simulated in a realistic environment (explicit solvent, ions, controlled temperature, physical forces).

The simulation tests whether the predicted binding mode is stable over time, rather than only geometrically plausible.

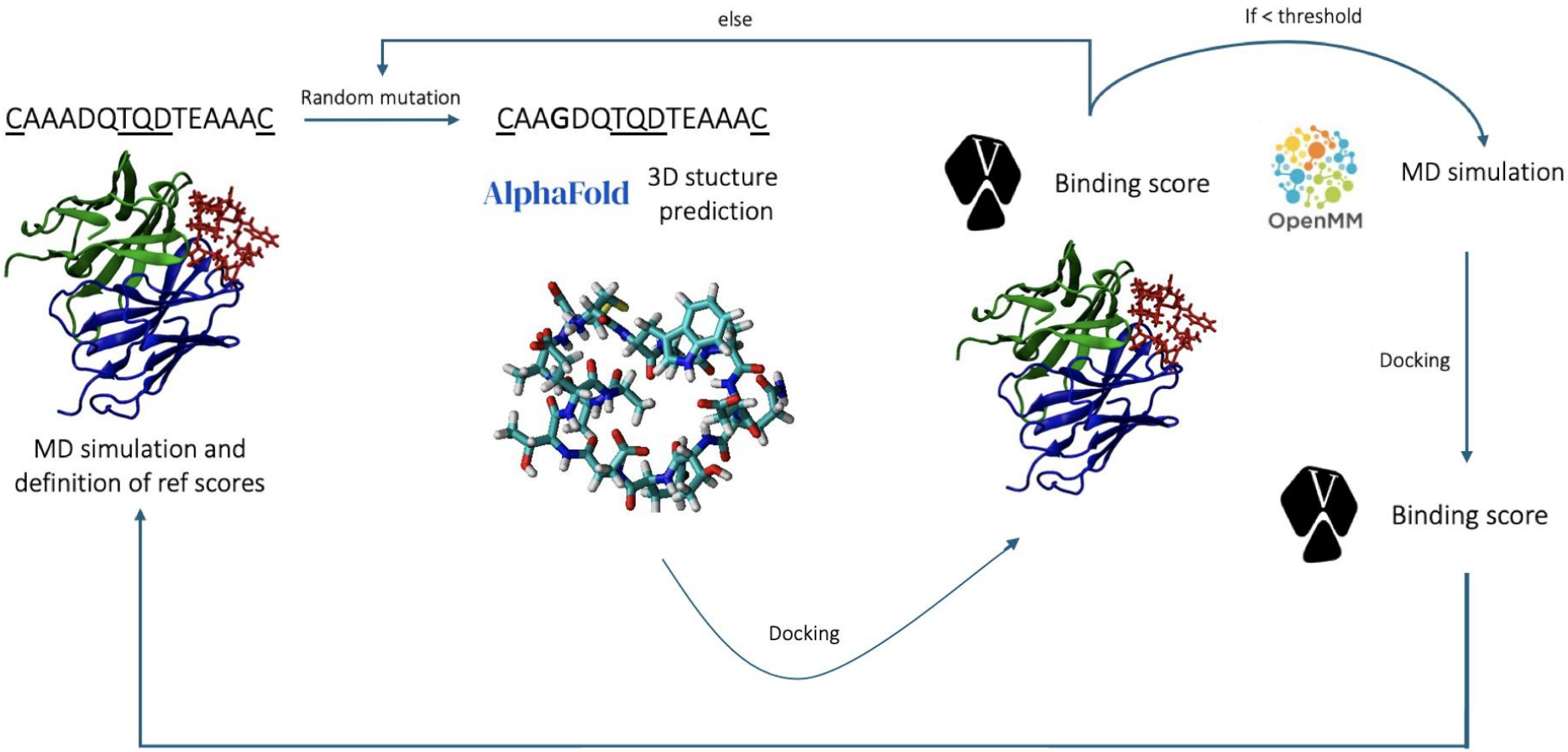
Docking results are filtered out by observing:

- binding persistence
- stability of key interactions
- structural integrity of the cyclic peptide

Role in the workflow: filtering and refinement of docking results under realistic conditions.

Optimization of existing peptide binders

MONTE CARLO ALOGRITHM



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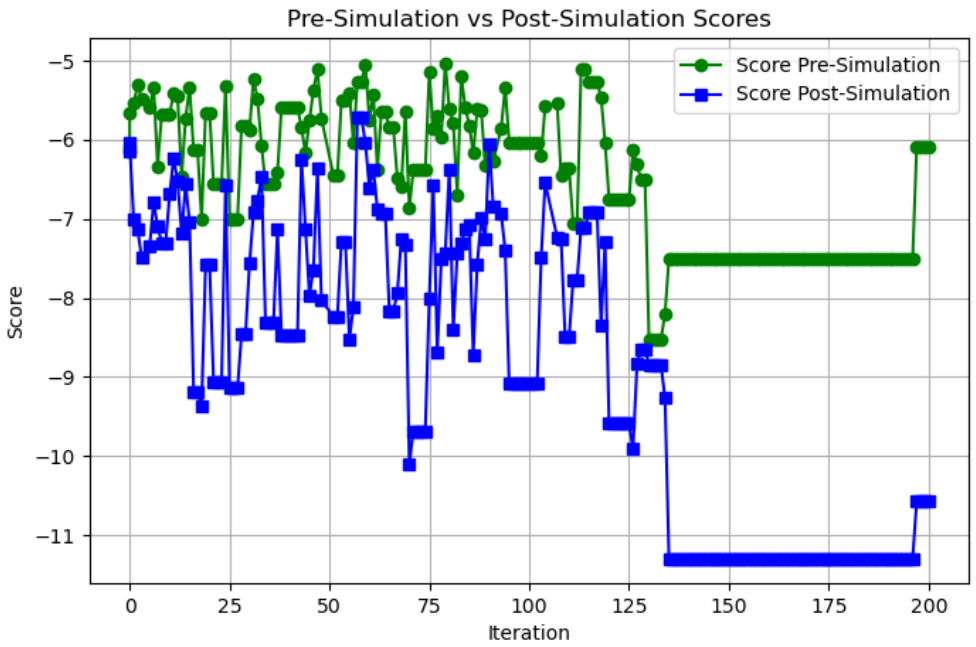
METROPOLIS CRITERION



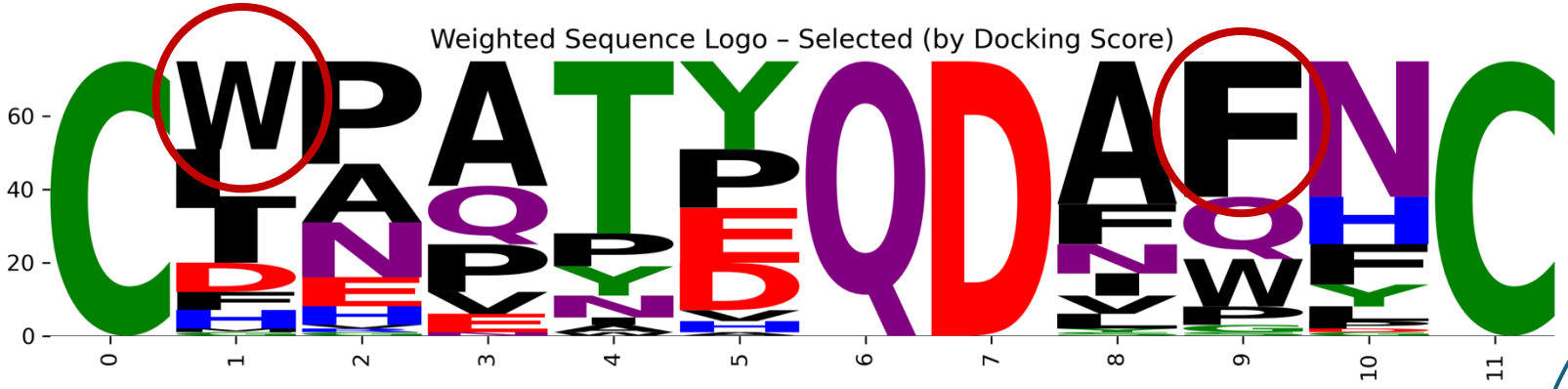

```
T = current_temperature                    # Cooling schedule-dependent
Pacc = min(1, np.exp(-delta_e / T))        # Acceptance probability
x = rng.uniform(0, 1)                      # Random number in [0, 1)
if delta_e < 0 or (x <= Pacc):
    # Accept mutation
else:
    # Reject mutation
```

If the new solution is better (delta_e < 0), it is always accepted.

If the new solution is worse (delta_e > 0), it may still be accepted with probability Pacc.

Allows acceptance of energetically unfavorable moves with a defined probability, preventing trapping in local minima, thus balancing exploration and optimization.

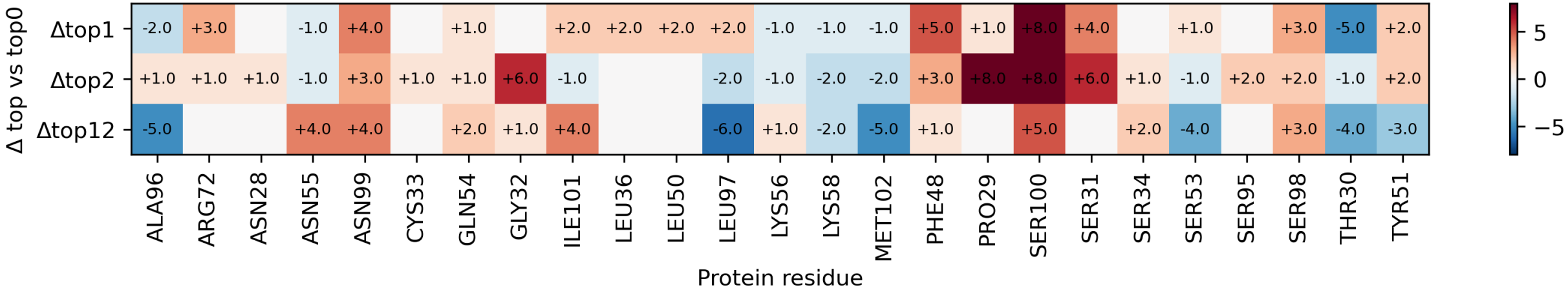
Optimization of existing peptide binders



Key interactions:

Top1: TRP2, PHE10

Top2: PHE10, THR5



Optimization of existing peptide binders

TOP_0

Residue	Num_Bonds
CYS1	4
ALA2	5
ALA3	12
ASP4	8
GLN5	3
THR6	6
GLN7	10
ASP8	2
ALA12	1
CYS13	6

TOP_1

Residue	Num_Bonds
CYS1	7
TRP2	6
PRO3	6
THR5	1
TYR6	12
GLN7	2
ALA9	6
PHE10	22
ASN11	8
CYS12	3

TOP_2

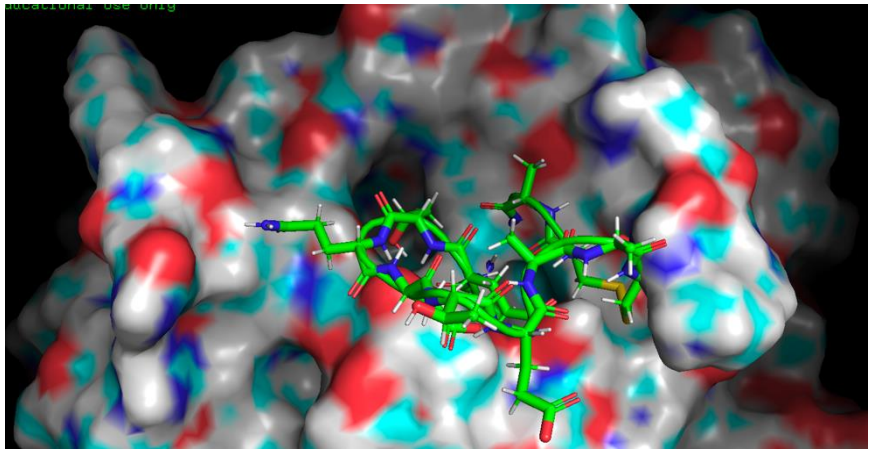
Residue	Num_Bonds
CYS1	2
ILE2	7
ASN3	10
GLN4	3
THR5	8
GLU6	11
GLN7	9
ASP8	1
ALA9	6
PHE10	17
ASN11	7
CYS12	1

TOP_12

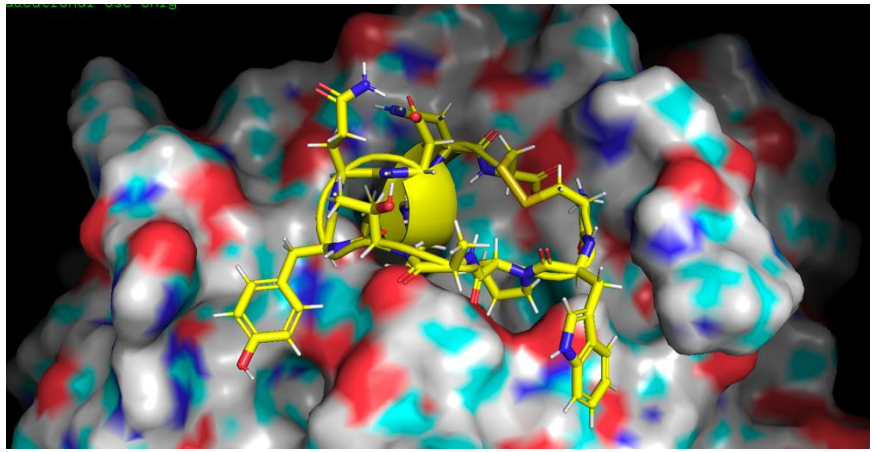
Residue	Num_Bonds
CYS1	3
HIS2	9
GLY3	5
GLU4	15
TRP5	10
TYR6	2
GLN7	4
PRO10	1
GLY11	3
CYS12	3

Optimization of existing peptide binders

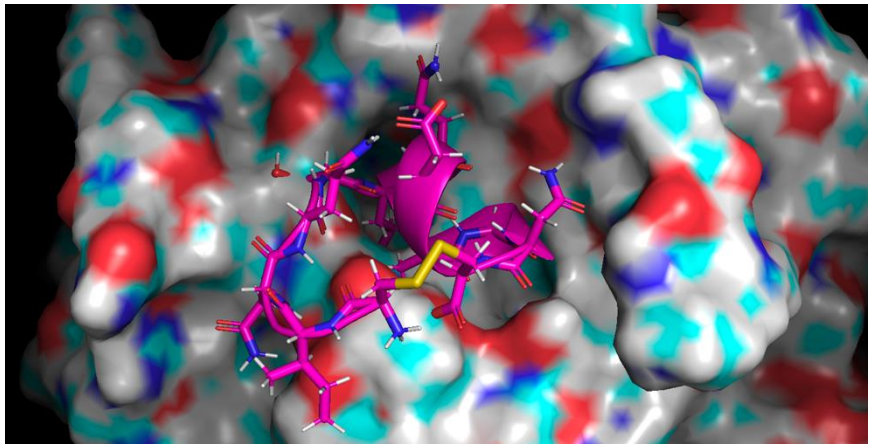
TOP0: -8.634 kcal/mol



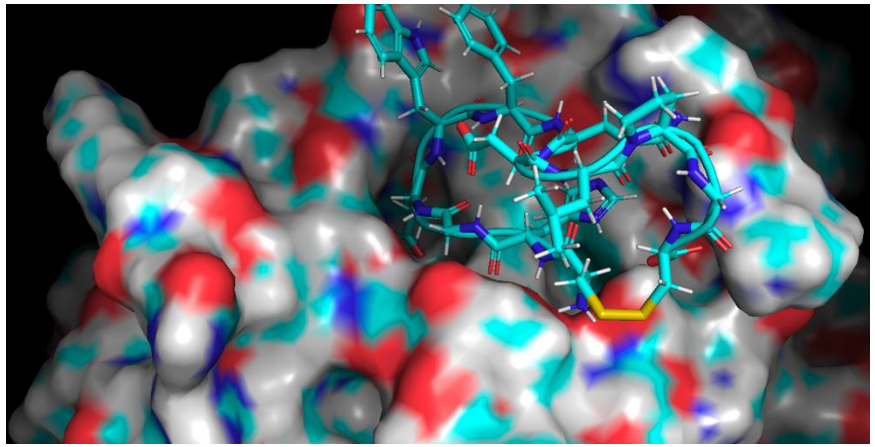
TOP1: -11.293 kcal/mol



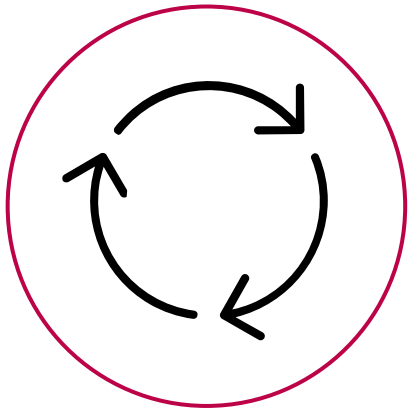
TOP2: -10.575 kcal/mol



TOP12: -9.061 kcal/mol

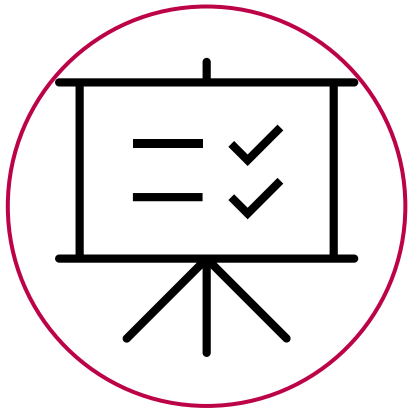


Optimization of existing peptide binders



Optimization of clinically relevant cyclic peptides with the desired properties

Faster than traditional physics-based methods



More reliable than Machine Learning-only algorithms

FUTURE PLAN: Using HPC systems to improve the scalability of the method

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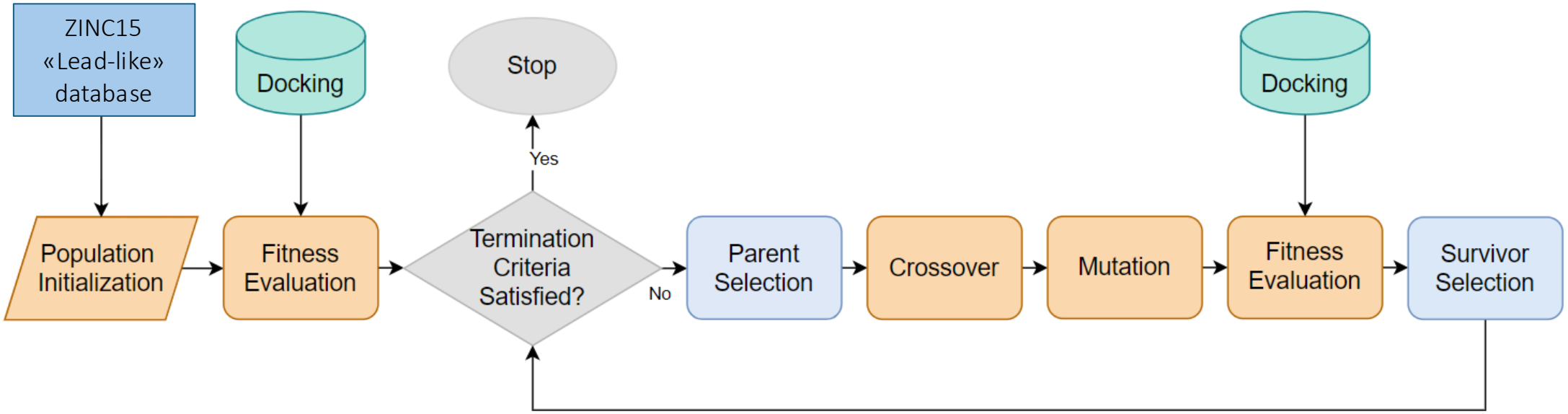
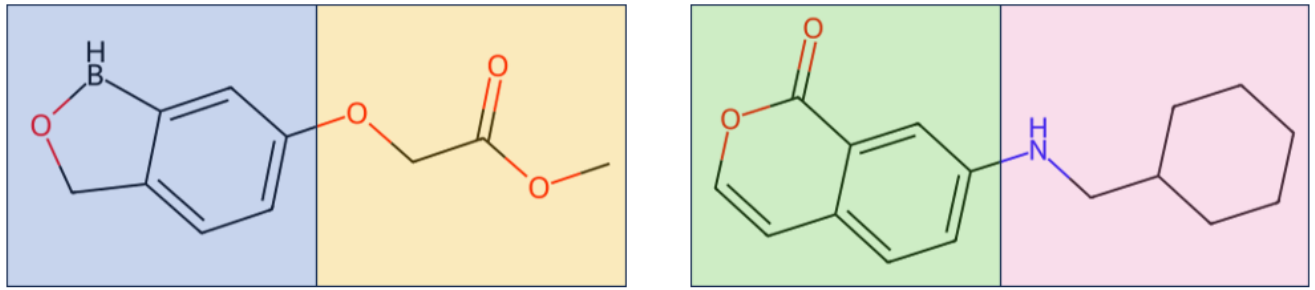
Evolution of molecules with a Genetic Algorithm



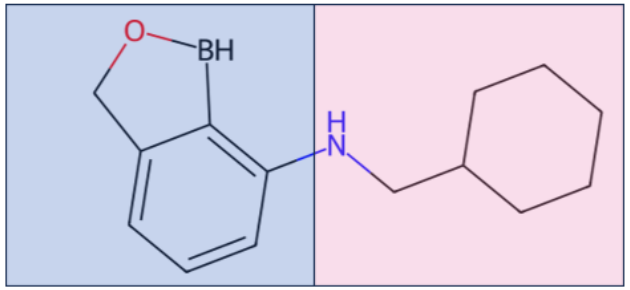
Evolution of molecules with a Genetic Algorithm



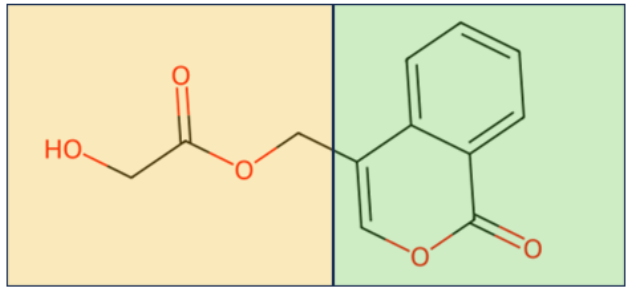
PARENT 1
COC(=O)COc1ccc2COBc2c1

PARENT 2
c1oc(=O)c2cc(NCC3CCCCC3)ccc2c1

Crossover



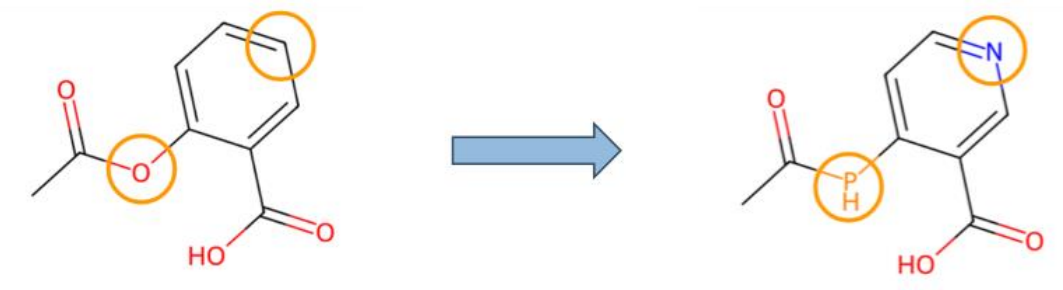
OFFSPRING 1
B1OCc2cccc(NCC3CCCCC3)c12



OFFSPRING 2
OCC(=O)OCc1coc(=O)c2ccccc12

Evolution of molecules with a Genetic Algorithm

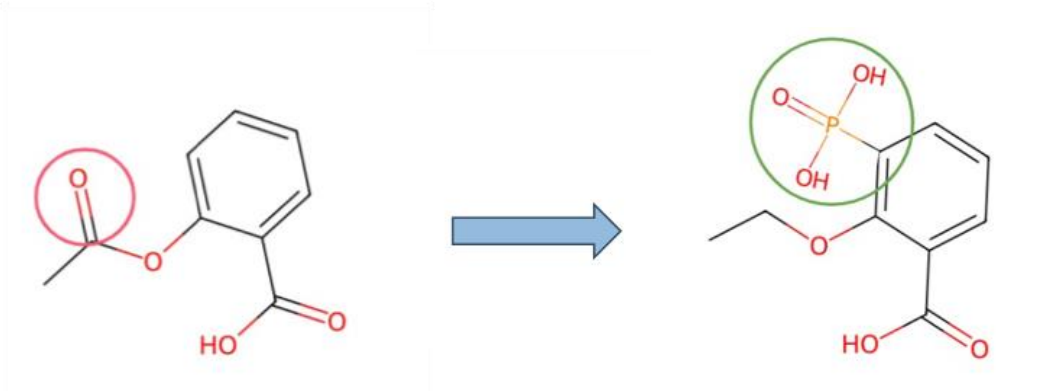
Atom-based mutation



CC(=O)Oc1ccccc1C(O)=O

CC(=O)Pc1ccncc1C(O)=O

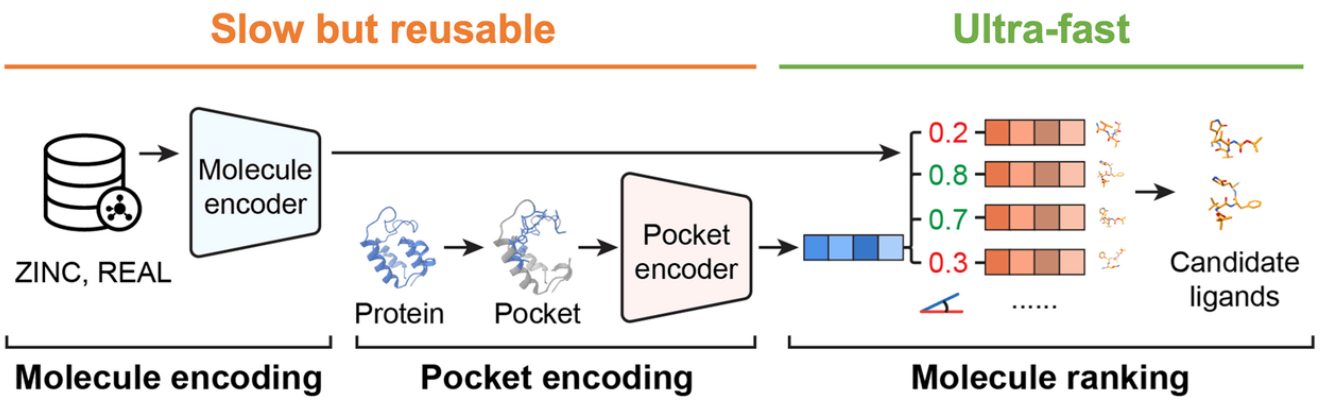
Fragment-based mutation

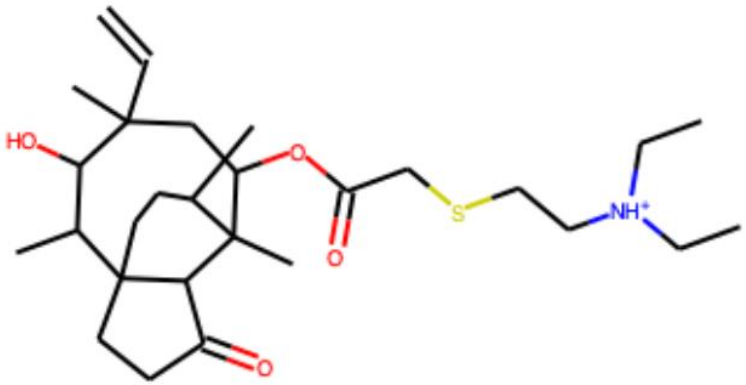


CC(=O)Oc1ccccc1C(O)=O

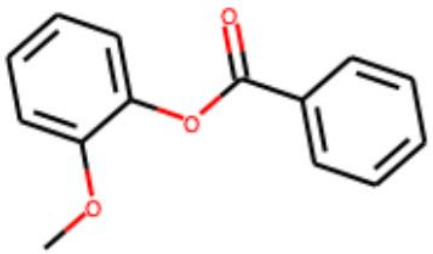
CCOc1c(P(=O)(O)O)cccc1C(O)=O

Evolution of molecules with a Genetic Algorithm





SA-score: 6.20



SA-score: 1.34

Drug Clip maps molecules and targets into the same embedding space → contrastive learning to understand which molecules are related to a target.

Similar molecules and targets are close in embedding space.

SA scorer (RDKit) measures how easy it is to synthesize a molecule (easier, 1 to harder, 10).

Evolution of molecules with a Genetic Algorithm

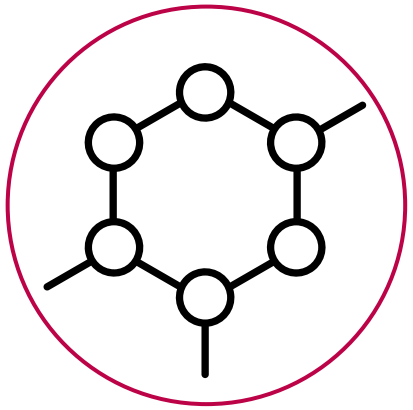
Preliminary results

gen	smiles	affinity	sa
189	NC(=O)Oc1npccc1C(=O)Nc1ccc(C(N)=O)cn1	0.8243	3.2320
93	NC(=O)OC1=NP=NC=C1C(=O)NC1=CC=C(C(N)=O)C=N1	0.8236	3.3148
168	NC(=O)OC1=NP=NC=C1C(=O)Nc1ccc(C(N)=O)cn1	0.8236	3.3148
78	NC(=O)C1=CN=C(NC(=O)C2=CC=PN=C2OC(=P)P)N=P1	0.8139	5.3165
96	NC(=O)C1=PN=C(NC(=O)C2=CC=PN=C2OC(=O)Br)N=C1	0.8072	4.5302
90	CC(=O)Oc1npccc1C(=O)Nc1ccc(C(N)=O)cn1	0.8038	3.1444
145	CC(=O)OC1=NP=CC=C1C(=O)Nc1ccc(C(N)=O)cn1	0.8038	3.1444
69	COOCOC1=NP=NC=C1C(=O)NC1=NP=C(C(N)=O)C=N1	0.7984	4.6012
132	NC(=O)OC1=NP=CC=C1C(=O)NC1=CN=C(C(N)=O)C=N1	0.7597	3.4581
36	NC(=O)C1=CN=C(NC(=O)C2=CC=PN=C2O)C=C1	0.7582	3.2497

Future plan

- Molecular dynamics
- Contacts analysis
- Scoring

Evolution of molecules with a Genetic Algorithm



Guided evolution of clinically relevant small molecules with the desired properties

Faster than traditional docking-based methods

Enables high-troughtput virtual screening of promising drug candidates

ACKNOWLEDGEMENTS

ALLIANCE PROJECT

RNA & GENETHERAPY

RESOURCES

LEONARDO CINECA

ISCRA-C application accepted

VENETIARUM UNIVERSITAS IN DOMO FOSCARI

VHPC

AND THANK YOU FOR YOUR ATTENTION